# MOHAMED HOSSAM

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## SUMMARY

Performance-driven Backend Developer specializing in building secure, production-ready web systems using Python. Expert in implementing Service Layer Architecture and database optimization to build maintainable, scalable applications. Currently preparing for GSoC 2026 to contribute to open-source ecosystems with a focus on high-concurrency backend systems.

## TECHNICAL SKILLS

**Backend & Architecture: Django, Django REST Framework, FastAPI, Service Layer Architecture, System Design, RESTful APIs, Asynchronous Programming.**

**Security & Auth: JWT/OAuth2, Session-based Auth, RBAC, CSRF/XSS Mitigation, Signed URLs (Cloudinary), Resource Hardening.**

**Databases: PostgreSQL , MySQL,, Redis, Schema Design.**

**DevOps & Tools: Docker (Multi-stage builds), Nginx, Gunicorn, Uvicorn, Linux CLI, Git, Postman, CI/CD .**

**Languages: Python (Expert), JavaScript (ES6+), C++ (Expert), SQL.**

## TECHNICAL PROJECTS

**CourseHub – Scalable Course Platform**

Django, PostgreSQL, HTMX, Cloudinary | GitHub| 2025 – Present

- Designed and developed a production-ready course platform capable of serving 10,000+ students with secure, session-based access control.
- Integrated Cloudinary CDNCloudinary CDN for scalable media delivery, including private video streaming with signed URLs, adaptive bitrate streaming, and automatic image optimization (WebP/AVIF).
- Built a layered backend architecture (Views → Service Layer → ORM) to isolate business logic and improve maintainability.
- Implemented passwordless email authentication using UUID-based verification tokens and otp with expiration handling.
- Designed a normalized PostgreSQL schema with indexed lookup fields to optimize concurrent access.
- Containerized the application using Docker and deployed with Gunicorn behind Nginx.

**FastAPI OCR Microservice**

FastAPI, Tesseract, Docker, Pydantic v2 | GitHub |2026 – Present

- Engineered a high-performance, stateless OCR microservice using non-blocking I/O.
- Implemented Resource Hardening including 10MB file limit and 30s timeout.
- Developed structured JSON logging and health/readiness probes.
- Architected containerized stack using Docker Compose and Nginx, ensuring horizontal scalability and environment parity.

**Clinic Management System (CMS)**

Django, PostgreSQL|GitHub|2025present

**Engineered** a comprehensive web-based dental practice management system for handling patients, visits, payments, and medical records in a unified workflow.
**Implemented** secure Role-Based Access Control (RBAC) using Django authentication to restrict access to sensitive healthcare data.
**Designed** a normalized relational PostgreSQL schema supporting patients, visits, payments, and X-ray records with optimized data relationships.
**Built** end-to-end patient lifecycle management including registration, visit scheduling, treatment documentation, and payment tracking.
**Developed** automated outstanding balance calculations and real-time payment status updates to improve financial transparency.
**Integrated** secure medical image handling with organized X-ray uploads, visit linking, and protected media storage.

### Chess Game

Python, Pygame | GitHub |2025 Present
**Developed** a modular chess game with full move validation, check/checkmate detection, and pawn promotion.
**Implemented** turn-based gameplay with a graphical board interface and game state handling.

## EDUCATION

**B.Sc. in Computer Science | Nahda University, Beni Suef | GPA: 3.23/4.0 |Expected Graduation: 2028**

## CERTIFICATIONS

**ALX Professional Foundations**
**ALX Ventures Freelance Academy**
**ICPC ECPC Qualifications – Honorable Mention**

## Achievements

**Ranked 20th** in **Constructor Open Cup 2026** – annual international programming competition organized by Constructor University & JetBrains Foundation.
**Solved 1100+ algorithmic problems** across Codeforces, LeetCode, and VJudge.
**Achieved Expert** rating on Codeforces.

## ADDITIONAL INFORMATION

**Languages: Arabic (Native), English (Fluent – B2).**